

Bias/Variance Tradeoff Notes

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Sampling Distribution of the Sample Mean

- $Y_i \sim_{iid}$ with $E[Y_i] = \mu$ and $V[Y_i] = \sigma^2$.
- Describe as a sampling problem and draw table and distribution
- Derive $E[\bar{Y}]$, what does this mean in terms of bias?
- Derive $V[\bar{Y}]$, how does this relate to $V[Y_i]$?
- Draw sampling distribution on top of distribution of an observation.

Sampling Distribution of the Sample Standard Deviation

- Put $S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$ on the board and ask what it is
- Derive $E[S^2]$,

$$\begin{aligned}(n-1) \cdot S^2 &= \sum_{i=1}^n (Y_i - \bar{Y})^2 \\&= \sum_{i=1}^n Y_i^2 - 2 \sum_{i=1}^n Y_i \cdot \bar{Y} + \sum_{i=1}^n \bar{Y}^2 \\&= \sum_{i=1}^n Y_i^2 - 2\bar{Y} \sum_{i=1}^n Y_i + \sum_{i=1}^n \bar{Y}^2 \\&= \sum_{i=1}^n Y_i^2 - 2\bar{Y}n\bar{Y} + n\bar{Y}^2 \\&= \sum_{i=1}^n Y_i^2 - n\bar{Y}^2\end{aligned}$$

$$\begin{aligned}
E[(n-1) \cdot S^2] &= (n-1)E[(n-1)S^2] \\
&= \sum_{i=1}^n E[Y_i^2] - nE[\bar{Y}^2] \\
&= nE[Y_i^2] - nE[\bar{Y}^2]
\end{aligned}$$

$$\begin{aligned}
E[Y_i^2] &= V[Y_i] + E[Y_i]^2 \\
&= \sigma^2 + \mu^2 \\
E[\bar{Y}^2] &= \sigma^2/n + \mu^2 \\
E[Y_i^2] - E[\bar{Y}^2] &= \sigma^2 - \sigma^2/n \\
&= \frac{n-1}{n}\sigma^2
\end{aligned}$$

- what does this mean in terms of bias? Draw sampling distribution.

Sampling Distribution with $\frac{1}{n}$

- Put $\tilde{S}^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2$ on the board.
- Derive $E[\tilde{S}^2]$,

$$\begin{aligned}
E[\tilde{S}^2] &= E\left[\frac{n-1}{n}S^2\right] \\
&= \frac{n-1}{n}E[S^2] \\
&= \frac{n-1}{n}\sigma^2
\end{aligned}$$

- what does this mean in terms of bias?
- write the $V[\tilde{S}^2]$ in terms of $V[S^2]$

$$\begin{aligned}
V[\tilde{S}^2] &= V\left[\frac{n-1}{n}S^2\right] \\
&= \left(\frac{n-1}{n}\right)^2 V[S^2]
\end{aligned}$$

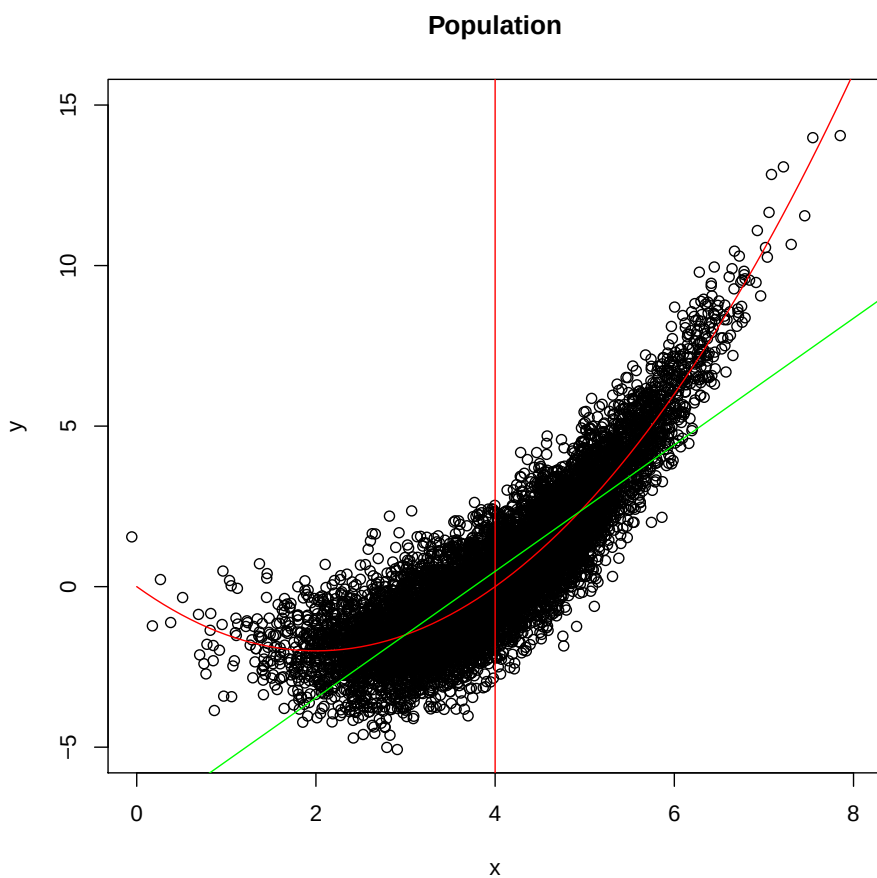
- draw sampling distribution for $\tilde{S}^2$ on top of sampling distribution for S^2 .
- discuss bias/variance tradeoff implicit in the comparison
- discuss $MSE = Bias^2 + Var$

Regression

```
# Population
N <- 10000
x <- rnorm(N,mean=4,sd=1)
y <- -2*x + .5*x^2 + rnorm(N)

plot(x,y,xlim=c(0,8),ylim=c(-5,15),main="Population")
lines(seq(0,8,.1),-2*seq(0,8,.1) + .5*seq(0,8,.1)^2, col="red")
abline(v=4,col="red")
abline(lm(y~x),col="green")

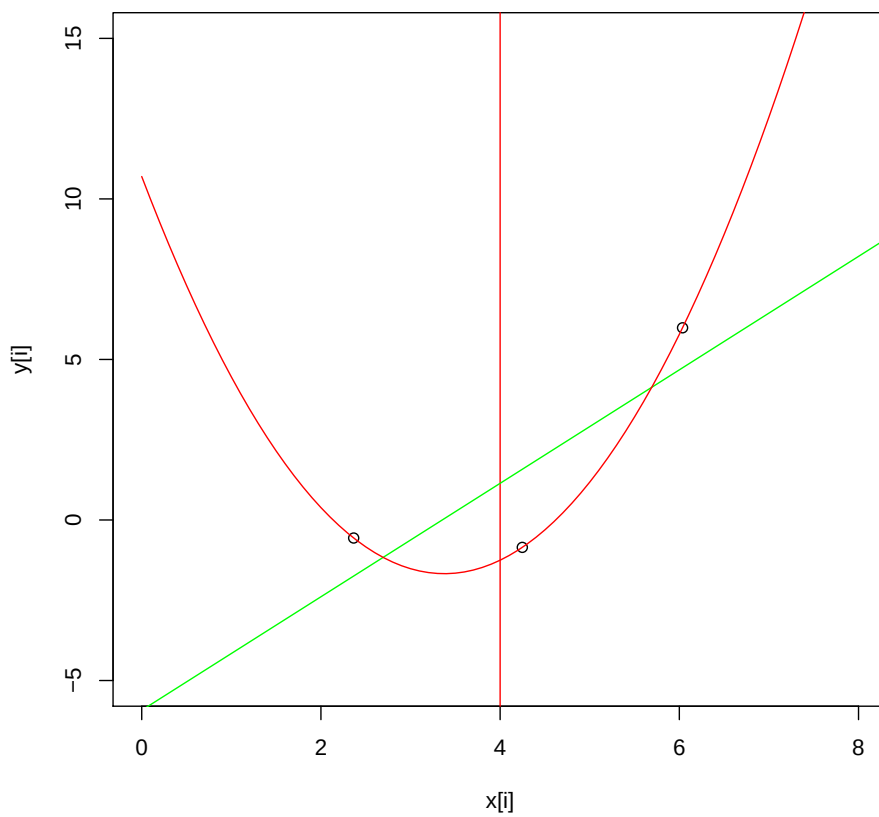
# f(4) = 0
-2*4 + .5*4^2
```



```

# Observed Data
set.seed(123)
n <- 3
i <- sample(1:N,size=n,replace = TRUE)
plot(x[i],y[i],xlim=c(0,8),ylim=c(-5,15))
abline(v=4,col="red")
abline(lm(y[i]~x[i]),col="green")
quad <- lm(y[i]~x[i]+I(x[i]^2))
lines(seq(0,8,.1),quad$coef[1] + quad$coef[2]*seq(0,8,.1) + quad$coef[3]*seq(0,8,.1)^2, col="red")

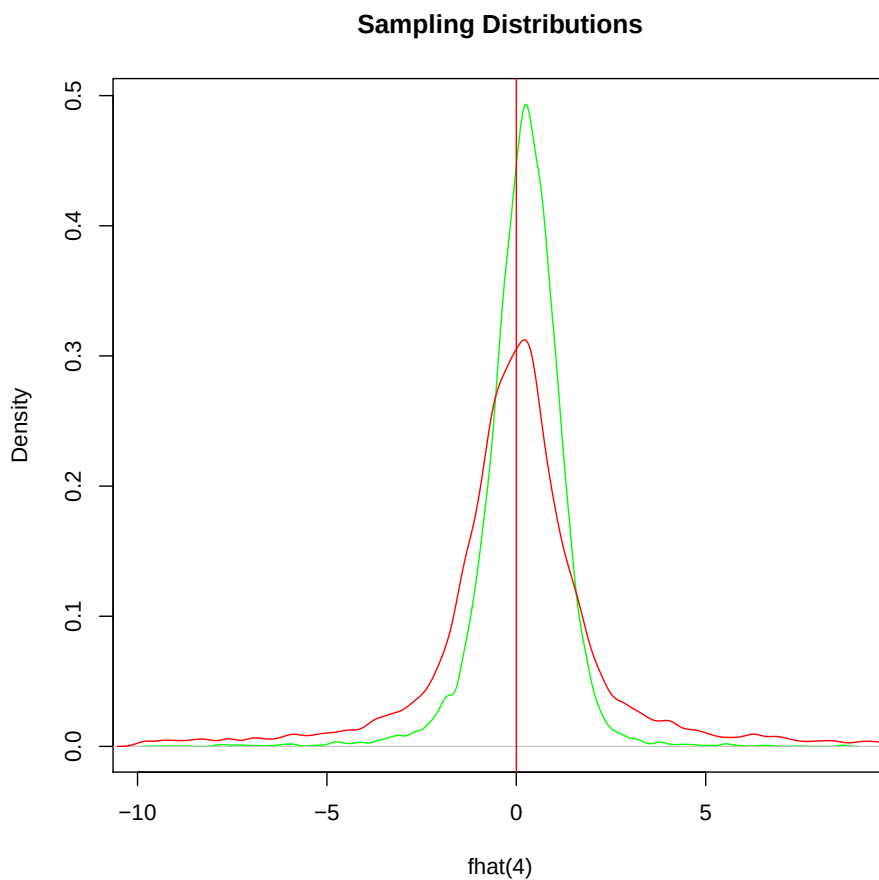
```



```

# Sampling Distributions
S <- 10000
fhat.lin <- rep(0,S)
fhat.quad <- rep(0,S)
n <- 3
for(s in 1:S){
  i <- sample(1:N,size=3,replace = TRUE)
  lin <- lm(y[i]~x[i])
  fhat.lin[s] <- lin$coef[1] + lin$coef[2]*4
  quad <- lm(y[i]~x[i]+I(x[i]^2))
  fhat.quad[s] <- quad$coef[1] + quad$coef[2]*4 + quad$coef[3]*(4^2)
}
plot(density(fhat.lin[fhat.lin > -10 & fhat.lin < 10]),col="green",main="Sampling Distributions")
lines(density(fhat.quad[fhat.quad > -10 & fhat.quad < 10 & !is.na(fhat.quad)]),col="red")
abline(v=0,col="red")

```



```
> #Bias
> mean(fhat.lin[fhat.lin > -10 & fhat.lin < 10])
[1] 0.1908053
> mean(fhat.quad[fhat.quad > -10 & fhat.quad < 10],na.rm=TRUE)
[1] 0.004055125

> #Var
> var(fhat.lin[fhat.lin > -10 & fhat.lin < 10])
[1] 1.191539
> var(fhat.quad[fhat.quad > -10 & fhat.quad < 10],na.rm=TRUE)
[1] 5.279145
```